# Omar Muhammed Ali

**Product-Minded Software Engineer | Backend Developer**

Egypt (Open to Remote)

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## Professional Summary

Backend Engineer with 2+ years of professional experience building scalable, user-focused systems using Django, NestJS, and PostgreSQL. Proven track record across multi-vendor marketplaces, on-demand transportation and delivery platforms, CRM systems, and subscription products, delivering asynchronous task pipelines, real-time WebSocket systems, wallet and payment-gateway integrations, and multi-tenant architectures. Skilled at translating product requirements into clean, well-tested, maintainable services, with a strong focus on system design, performance optimization, and measurable business impact.

## Education

**Bachelor of Computer Science** **September 2023 – May 2027**

*Obour Institute for Management and Computer Technology* *Obour, Cairo, Egypt*

- Relevant coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Software Engineering, Computer Networks

## Technical Skills

**Programming Languages:** Python, JavaScript, TypeScript, SQL, HTML5/CSS3
**Backend Frameworks:** Django, Django REST Framework, NestJS, FastAPI, Node.js
**Software Engineering:** Data Structures & Algorithms, OOP, Design Patterns, Service-Layer Architecture, System Analysis & Design, TDD
**Databases & Messaging:** PostgreSQL, Redis, MySQL, SQLite, RabbitMQ, Celery
**Cloud & DevOps:** Docker, Docker Compose, Nginx, CI/CD, Git, Linux, VPS Management
**Web Technologies:** WebSockets (Django Channels), JWT, OAuth2, RESTful APIs, GraphQL, Swagger/OpenAPI
**Payments & Wallets:** Moyasar, PayMob, Stripe, PayPal, In-App Wallet Systems, Webhook Processing

## Professional Experience

**Software Engineer** **July 2026 – Present**

*Tradeway* *Mohandiseen, Egypt (Hybrid)*

- **CRM Architecture:** Currently architecting and developing a large-scale, comprehensive CRM system to streamline customer relationship management, optimize sales workflows, and handle complex data relationships.
- **Accounting Systems:** Refactored core backend accounting services to improve financial data processing reliability, wallet transaction integrity, and automated reporting.
- **Marketing Analytics:** Built multi-channel marketing analytics dashboards and data aggregation pipelines to track campaign performance and metrics in real-time.
- **Infrastructure & Database Management:** Implemented robust automated database backup strategies and utilized Docker alongside PostgreSQL and MySQL to ensure high availability and data integrity across multiple service environments.

**Backend Developer** **May 2026 – August 2026**

*EES – Multi-Vendor E-Commerce & Services Platform* *Saudi Arabia (Remote) | Contract*

- **Multi-Service Transportation Backend:** Designed and integrated REST APIs powering four distinct service verticals — taxi, delivery, bus, and specialized truck transportation — within a single multi-vendor platform.
- **Request Lifecycle Engine:** Built end-to-end customer and driver request management covering request creation, real-time status tracking, driver price bidding, ride cancellation, and completed-request history, ensuring consistent state transitions across all service types.
- **Driver & Fleet Management:** Developed full CRUD APIs for DriverCar management with automated test coverage, enabling reliable vehicle onboarding and driver-to-vehicle assignment.
- **Bus Operations:** Implemented bus journey scheduling and management APIs alongside customer ticket verification endpoints for boarding validation.
- **Wallet Payments:** Engineered a unified wallet payment-processing flow shared across multiple services, backed by supporting test suites to safeguard transaction integrity.
- **Architecture & Maintainability:** Refactored the transportation request architecture and reorganized taxi APIs to cleanly separate customer and driver operations, delivering serializers, data models, URL routing, service-layer workflows, and migrations with integration tests to reduce coupling and speed up feature delivery.

**Backend Engineer** **October 2025 – Present**

*Madinah Innovation* *Riyadh, Saudi Arabia (Remote) | Part-time*

- **Neutration (Fitness & Nutrition Platform):** Engineered an AI-powered fitness ecosystem with personalized meal generation using asynchronous background processing (Celery/Redis) and real-time specialist-client collaboration via WebSockets, reducing meal planning response time significantly.
- **Subscription & Monetization:** Built a robust payment flow integrating Moyasar API with webhook-driven logic for tiered subscriptions, quota management, and feature entitlements, supporting multiple subscription tiers.
- **UniApps Platform:** Architected a multi-tenant Django platform connecting university students with merchants via an automated voucher and offer distribution engine, serving multiple university campuses.
- **Gamification & Engagement:** Implemented user streaks, wallet rewards, and achievement tiers to drive long-term retention and measurable increases in daily active usage.
- **Infrastructure & Async Systems:** Optimized containerized deployments (Docker/Docker Compose) and developed scalable notification pipelines using Celery and FCM for real-time status updates.

**Backend Engineer** **July 2025 – October 2025**

*Functo Software House* *Cairo, Egypt (Remote) | Full-time*

- Refactored critical CRM components, increasing system throughput by **30%** and improving database query efficiency through targeted index optimization and query restructuring.
- Developed the "Buyer Match" engine — an automated property-matching algorithm using advanced PostgreSQL filtering logic, reducing manual search time for agents.
- Implemented "Unit Share" modules to handle complex partial-ownership transactions and fractional property management workflows.
- Optimized RBAC (Role-Based Access Control) security layers, reducing authentication overhead across distributed services.

## Featured Projects

**Twitter API Clone** | *Python, Django, WebSockets, Redis* | GitHub

- Built a real-time social platform with tweet management, custom recommendation algorithms, and live WebSocket notifications for follows, likes, and mentions.
- Implemented Redis-backed caching for timeline queries and Celery workers for async notification delivery, handling high-concurrency read patterns.

**E-Learning Management System** | *Django, PostgreSQL, Docker, JavaScript* | GitHub

- Developed a full-featured LMS with structured course hierarchies, student progress tracking, and instructor dashboards supporting multiple content types.
- Containerized the full stack with Docker Compose and implemented role-based access for students, instructors, and admins with granular content permissions.

**E-Commerce Platform API** | *Django, Celery, Stripe, Redis* | GitHub

- Architected a scalable REST API supporting high-concurrency transactions with integrated Stripe payment gateway, inventory management, and order lifecycle tracking.
- Implemented async order processing with Celery, reducing checkout-to-confirmation latency and enabling background email/notification workflows.

## Additional Information

- **Languages:** Arabic (Native), English (B2 – Upper Intermediate)
- **Problem Solving:** Expert at breaking down business requirements into technical roadmaps and debugging complex production issues.
- **Collaboration:** Proficient in Agile/Scrum environments, bridging communication between technical and non-technical stakeholders.